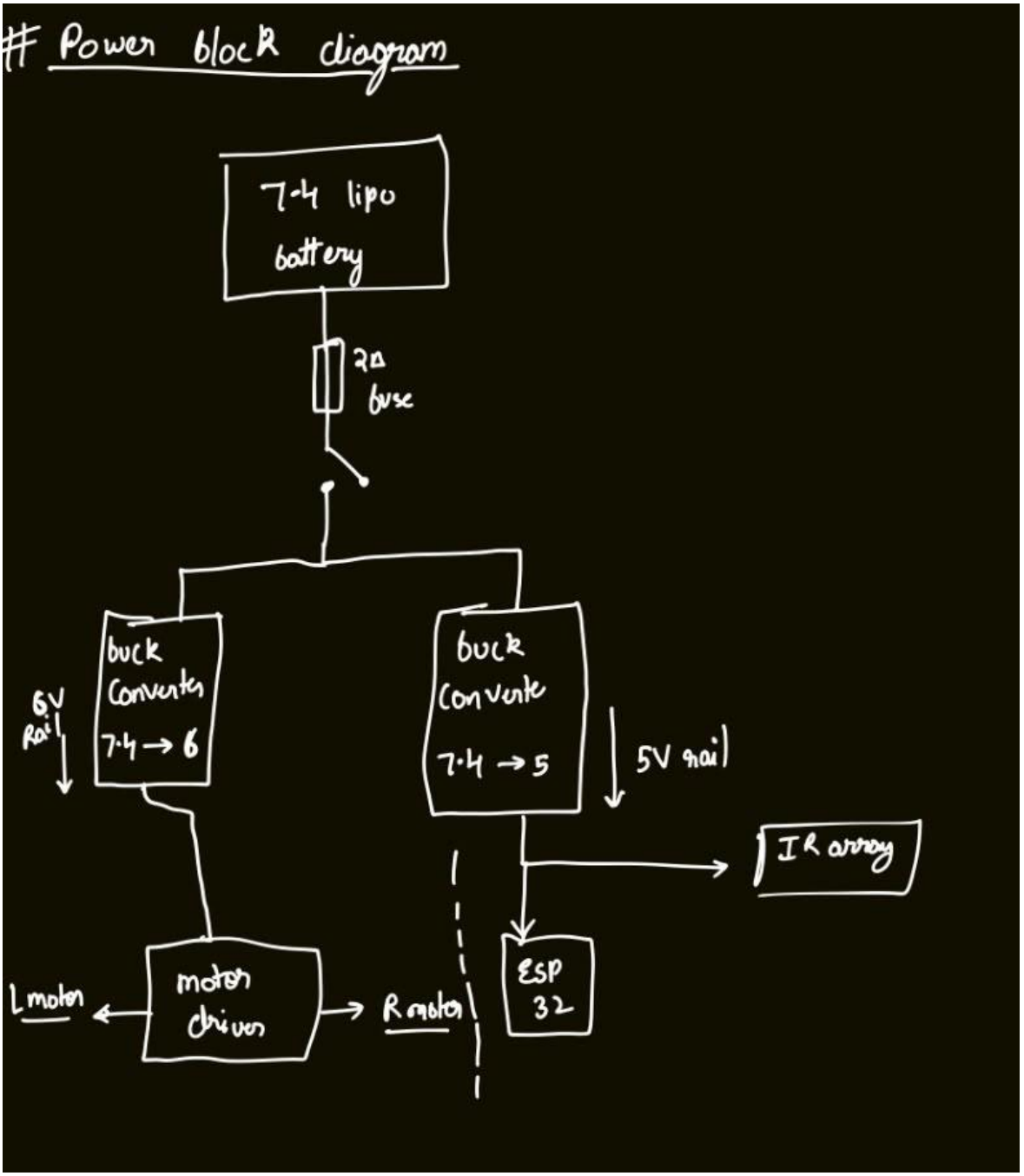
Task 5 - Power budget and battery selection

Power Block diagram



Battery selection and justification

- We can use the 2S LiPo · 7.4 V · 2200 mAh · 80C battery
  - The main reason why we use lipo batterys is that they deliver high current for their size ..so something like the compact bot of line maze solving ..
  - And also the motors need 6v so its easier to convert 7.4 to 6 and also to 5v basically

- Link - https://www.amazon.in/Original-ABSD-7-4V-2200MAH-Battery/dp/B0GCBRRP1Y?crid=3VTJ8JDSBHGPD&dib=eyJ2IjoiMSJ9.SsgH_-TyyLoZD2CLUlbU0bpgmFQODpvPgVIsCbn-h0va5gWtLmbxCz97PjToHbCnClyJOUIQvBcGzsP3wF2qcVOAWIdjM_Fok_t9YcMLL0V_rLos2MWrSsnzWMsbsHgdEVd-BumfBdXxhihwtLxkucYq3hu3HXNgWjYf-AmXcBezcLUmYJgl0VZ7VcbEvBhqzUsyVmVo_-haakVrdB8Dt3ToiqXtFjcq6AcMqViDUyg.gu161wE9vl6iEiQq6nRAd14dQ8ro1RZQvzNQ2JiEYNQ&dib_tag=se&keywords=2S+LiPo+%C2%B7+7.4+V+%C2%B7+1000+mAh+%C2%B7+25C&nsdOptOutParam=true&qid=1792959097&sprefix=2s+lipo+7.4+v+1000+mah+25c%2Caps%2C370&sr=8-3
- Price – 1600 Rs

Current draw estimation

- Lets talk abt the ratings first then go into calculations
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Component	Idle	Peak	Basis
ESP32 DevKit	~80 mA	~250 mA	The chip draws 20–68 mA depending on clock speed with the radio off, and 240 mA at Wi-Fi transmit peak. I added ~10 mA for the DevKit's USB-UART chip, LED and regulator (my estimate).
QTR-8RC array	~100 mA	100 mA	LED current is about 20–25 mA per string, giving just under 100 mA for the board. Pololu lists the supply current as 100 mA.
Left N20 motor (6 V)	60 mA	550 mA	One common encoder-equipped N20 lists 0.55 A stall and under 0.06 A no-load at 6 V. SparkFun's N20 pair lists 0.03 A no-load and 0.45 A stall. I used the higher figures.
Right N20 motor (6 V)	60 mA	550 mA	Same as above.
Encoders (both)	~10 mA	~20 mA	Vendors only state a 3.3 V or 5 V supply, not a current. I assumed 5–10 mA each for the Hall sensors and pull-ups.
TB6612FNG	1.5 mA	2.2 mA	Logic current is 1.1–1.5 mA typical and up to 2.2 mA max, and standby is 1 µA max.
Buck converters (×2)	~10 mA	see Step 3	The LM2596 has about 5 mA quiescent current and 80% efficiency at 5 V out, 12 V in, 3 A.

- Summin the currents for rails .. there wll be 2 rails ..5v and 6v rails so

Rail	Idle	Peak
5 V logic (ESP32, QTR, encoders, TB6612 logic)	~192 mA	~372 mA
6 V motors (2 × N20)	120 mA	1,100 mA

- The worst-case total is about 1.5 A (just add the peak currents across all rails)

- But The max current that the battery i selected can give abt 176 A so it might be so that (say in a short more current than the 1.5 A is drawn in fact very high current may be drawn) in that case to protect my comopnets i need a fuse so we need a 3A fuse ....

Runtime calculation

- We are using a 2s 2200 mAh 80c Battery .. so we cannot use 100% of the battery say we use 75% i.e 1650 mAh and coz of the 80c we get 80*2200mAh = 176 A so the battery can supply this much current basically
- Suppose the typical driving and everything works normally and there is not stall and shii so then power of 5v rail is (0.4 + 0.5 + 0.06 ~ 1W) and the 6v rail power is (0.36 + 0.36 ~ 0.72) so total approximately is 1.7W
- So since the battery is 2200 mAh and 7.4 the total energy it has is 2.2* 7.4 but we considering only 75% so 2.2* 7.4(0.75) = 4.7 so we draw approx 1.7 W ideally so approx the car should work for ~ 7h ideally

Cell chemistry justification

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Property	Li-ion (18650 cells)	LiPo (RC pouch pack)
Energy density	200–265 Wh/kg at cell level	150–200 Wh/kg at pack level
Max discharge rate	2–10C for standard cells	20–120C for RC-grade packs
Cycle life	500–1000 cycles	150–500 cycles
Self-discharge	1–3% per month	2–5% per month
Safety	The metal case makes it more robust to physical damage; LiPo has a medium rating because of puffing.	Soft pouch, so it can swell and be punctured. It usually has no built-in protection.
Shape	Rigid cylinder, needs a holder	Flat and low-profile, easy to fit in a chassis Table of Contents +8

- Why im using a LiPo battery compered to L ion battery
    - It's flat and compact, so it sits low on a small chassis, and you don't need a cell holder.
    - It handles motor start-up spikes and stalls with enormous headroom.
    - Its 7 h typical runtime is more than enough for a robot run.
- The disadvantages
    - **It's oversized.** My worst case is 1.44 A, which is 0.65C on a 2.2 Ah pack, while an 80C pack can deliver up to 176 A. The rating and extra capacity are mostly wasted, and they add weight.
    - **Developing on the big pack and swapping later is going to be my strategy(if the battrey i chose is that bulky then we can change to a smaller one after everything else works).** It gives long test sessions without recharging. It also has low internal resistance